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A possibility-based multi-criteria decision-making approach for artificial recharge structure selection using pentagonal fuzzy numbers

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ABSTRACT

Groundwater recharge is crucial in developing groundwater sources in an area for regional growth. One of the essential natural water resources is groundwater, which differs geographically in quality and amount. Groundwater resources are under great strain due to growing urbanization and population growth. Numerous studies have identified groundwater recharge regions using a geographic information system and Multi-Criteria Decision-Making (MCDM). In this study, we have proposed an Multi-Criteria Decision-Making technique for selecting artificial recharge structures in basaltic terrain. The proposed Multi-Criteria Decision-Making technique is built with possibility theory under a pentagonal fuzzy environment. Additionally, we have developed a possibilistic middle of asymmetric linear pentagonal fuzzy (LPF) number. Through the Multi-Criteria Decision-Making technique, we have selected the best artificial recharge structures from the available ones. To show our Multi-Criteria Decision-Making methods feasibility, we have considered one numerical example under a pentagonal fuzzy environment.

1. Introduction

All stages of life, like survival, are crucial for relieving poverty, hunger, and disease and critical for economic use in our daily life; water is the most important of the earth for our life and environment and plays the leading role in sustaining life on earth. The role of water is unique; nothing can substitute it. Safe drinking water is the primary motto of every population. Surface water is becoming scarce, and groundwater has emerged as an essential source to meet India's water requirements for various sectors, including irrigation (89%), domestic (9%), and industries (2%). Groundwater constitutes the largest share, about 61% for irrigation, followed by canals with 24%. 50% of urban water requirements and 85% of rural domestic water requirements are also fulfilled by groundwater. The last few decants, the rapid growth of populations and civilization, huge waste and uses of water, inefficient irrigation and pollution are creating water pressure. Water pollution or contamination is also playing major role in this regard. Water demands are increasing rapidly. Many parts of the world, even India, have been under pressure water. In India, groundwater is vital to fulfilling domestic, industrial, and agricultural demands. The increasing water demand in India is realized that the planned artificial recharge (AR) is needed to enhance the groundwater quality and quantity. AR helps protect surface runoff, prevent seawater intrusion [1], reduce overdraft, and increase available groundwater supplies and groundwater quality [2]. It impedes flood and minimizes the impact of the flood. It stores additional runoff water, particularly heavy rainfall, for future use. It also reduces flood peaks and damage and conserves the soil.

The main objective of our work is to find the best AR technique through the decision-making technique. AR techniques are a substantial development to develop the groundwater label. The main objectives of our articles are described as:

1. The work aims to select the best artificial groundwater recharge structure on the natural groundwater system, maintain the groundwater level & manage the groundwater.
2. To preserve the additional surface water for future uses, particularly during the rainy season, and reduce the flood peaks.
3. To recharge the groundwater in the basaltic terrain.
4. To improve the water quality. When rainwater passes through the soil during groundwater recharge, the soil acts natural filter.

Few workers have workers on artificial recharge with decision-making approaches. A few techniques were used for more precise groundwater potential zone mapping in various areas around the world,

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Table 1
Comparing our MCDM approach with reference methods.

Authors	Decision making methods	Environment	Application
Lin et al. [13]	TOPSIS method	Hesitant fuzzy linguistic term sets	MCDM problem
Bajaj & Kumar [22]	Correlation coefficient approach	Intuitionistic fuzzy	MCDM problem
Jana et al. [14]	Dombi-aggregation functions	Fuzzy	MCDM problem
Chakraborty et al. [31]	Ranking & Defuzzification	Pentagonal fuzzy	Game problems
Wan & Dong [27]	Possibility method	Intuitionistic fuzzy	Multi-attribute group decision making problem
Yilmaz & Harmancioglu [32]	MCDM method	Crisp	Water resource management
Panda & Pal [33]	Pentagonal fuzzy matrix	Pentagonal fuzzy	Multi-criteria decision making
Our article	Possibilistic middle MCDM method	Symmetric pentagonal fuzzy	Artificial recharge structure selection

including multi-criteria decision-making (MCDM), probabilistic models (PM), multivariate statistics (MS), bivariate statistics (BS), and data-mining/machine learning (DM/ML). MCDM is a useful tool for making decisions with too many influencing variables [3]. Abijith et al. [4] studied multi-criteria decision analysis (MCDA) using geographic information systems (GIS) to find possible groundwater recharge zones. Managed aquifer recharge is applied using spreading techniques based on the GIS-MCDA approach [5]. The application of GIS-MCDA in mapping the potential for groundwater recharge was found by Maghribi et al. [6]. Geo spatial data representing the research area are combined and given weight following the research goals to conduct the location selection process [7]. Kumari et al. [8] identified the conceivable groundwater recharge zone(s) using GIS-based MCDM and sub-surface mapping from the Electrical Resistivity Tomography (ERT) procedure.

Uncertainty is frequently present when using the MCDM technique to solve water resource management issues. There may occasionally be a need to provide information regarding the MCDM problem that is more consistent, comprehensive, and clear. Fuzzy sets (FSs) are a notion that Zadeh [9] created to deal with ambiguity in real-world problems. In the decision-making problem, so much information happens together; therefore, a fuzzy environment is very useful for this case. Hutton et al. [10] deal with the uncertainty of water distribution system models. Luo and Jennings [11] study the MADM problem under the fuzzy environment. A novel fuzzy decision-making methodology for ranking energy storage investments was investigate by Oflaz et al. [12]. Many researchers work on decision-making under uncertain environments like: Lin et al. [13], Jana et al. [14], Singh et al. [15], Garai and Garg [16], Tian et al. [17], Oyelade et al. [18], Abualigah et al. [19], Ezugwu et al. [20], Garai and Garg [21] and Bajaj and Kumar [22].

One uncertainty theory in FSs that Zadeh [23] developed is the possibility. In FS theory, both its practical application and theoretical idea are important. Because of time constraints, many actual decision-making problems include imperfect facts. Consequently, the possibility measure was used by Dubois and Prade [24] to describe the fuzzy expectation of interval fuzzy numbers. It was suggested by Carlsson and Fuller [25] that a fuzzy number may have a mean and variance. The definitions of the weighted upper and lower possibilistic means were provided by Fuller and Majlender [26]. The idea of mean and variance of triangular intuitionistic fuzzy numbers was suggested by Wan et al. [27]. The concept of possible mean, variance, and covariance of a generalized intuitionistic fuzzy number was introduced by Garai et al. [28]. Garai et al. [29] suggested a ranking system for neutrosophic numbers based on the possibility mean. The weighted possibility means of fuzzy are defined by Dong and Wan [30]. However, we have presented what is the difference between our work and others in Table 1.

MCDM is a technique for choosing several alternatives assessed according to predetermined criteria. Spatial research is typically used to decide whether a location is suitable. Few researchers work on AR with decision-making approaches. Maghribi et al. [6] figured out how to identify potential groundwater recharge areas, mainly using the multi-criteria GIS technique. In the recent decay, many researchers applied the MCDM technique to different water resource management systems, like as: [32,34–38]. However, no one studied the MCDM technique on Artificial Recharge under uncertain environments. Therefore in this study, we have to select of Artificial Recharge structure in basaltic terrain under a pentagonal fuzzy environment.

1.1. Motivation

This research develops an artificial recharge structure model to make indicator-based decisions concerning environmental, social, and economic dimensions from a multiple-criteria viewpoint. In the proposed, the Deccan trap is dominated by hard igneous rock and is generally considered a poor aquifer. Deccan trap, a Semi-arid climate with very insufficient rainfall, gives rise to an acute shortage of groundwater and does not receive significant rainfall. This low rainfall does not support wide-ranging and water-intensive agriculture in the region. In the Deccan trap area, most of the cultivable lands are dependent on irrigation from groundwater. The groundwater potential zones are mainly observed in pediplains with weathered basaltic rocks and low drainage density. To availability, the groundwater level is very turf job. Groundwater availability depends on this trap's geology, soil type, geomorphology, land use and land cover, climate, and topography. Many uncertain factors exist when we try to increase the groundwater label with the help of artificial recharge. Therefore, in this article, first, we developed one ranking method with this idea to build up a novel MCDM technique. Due to many uncertain factors, we are in a fuzzy environment, and we can easily identify the best artificial research technique. So, our research technique is beneficial for our society.

1.2. Implementation of this work

This paper compiles a critical analysis of geology and basaltic terrain regarding groundwater recharge. This will lead the young researcher to understand the complete scenario of the basaltic rock, especially in Deccan trap regions in India. We have also discussed existing AR methods mainly used in India's peninsular. Throughout this research, we have developed the following new idea.

- Defined the possibilistic middle of an asymmetric linear pentagonal fuzzy number.
- Developed the MCDM technique based on the possibilistic middle.
- Application of possibilistic middle-based MCDM on Artificial Recharge structures specific in Basaltic Terrain.
- Numerically solved Artificial Recharge structures specific in Basaltic Terrain decision-making problem under LPF-environment.

1.3. Structure of the paper

The rest of the paper is organized as follows: In Section 2, we have discussed the basic concept of LPF-number and its basic properties. Possibilistic middle and novel ranking method of LPF-number defined in Section 3. The possibility of a middle-based MCDM technique is studied in Section 4. In Section 5, we proposed MCDM on Artificial Recharge structures specific in Basaltic Terrain and its numerical illustration. We have discussed the advantages of our proposed method in Section 6. The conclusion and future work plan are studied in Section 7.

2. Basic preliminaries

Definition 1. A fuzzy number (FN) $\tilde{A}$ is a subset of the real line $\mathbb{R}$, and the membership function $\mu_{\tilde{A}}$ has the [33] following properties:

(i) $\mu_{\tilde{A}}$ is a convex set, i.e., $\mu_{\tilde{A}}(\lambda x_1 + (1-\lambda)x_2) \geq \min(\mu_{\tilde{A}}(x_1), \mu_{\tilde{A}}(x_2))$ for all $x_1, x_2 \in \mathbb{R}$.
(ii) $\tilde{A}$ is satisfy the normal property, i.e., $\exists\, x_1 \in \mathbb{R}$ such that $\mu_{\tilde{A}}(x_1) = 1$.
(iii) In its domain, $\mu_{\tilde{A}}(x)$ is piecewise continuous.
(iv) The support $\tilde{A}$ is bounded.

Definition 2. A FN is said to be $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5\ddot{e}$, $p_1 \leq p_2 \leq p_3 \leq p_4 \leq p_5$ is said to be pentagonal fuzzy (PF) number [33], if the membership function ($\mu_{\tilde{a}}(x)$) satisfy the following conditions:

(i) $\mu_{\tilde{a}}(x)$ is continuous in the region $[0, 1]$.
(ii) $\mu_{\tilde{a}}(x)$ is strictly increasing in the interval $[p_1, p_2]$ & $[p_2, p_3]$.
(iii) $\mu_{\tilde{a}}(x)$ is strictly decreasing in the interval $[p_3, p_4]$ & $[p_4, p_5]$.

Definition 3. Two PF-numbers $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5\ddot{e}$ and $\tilde{q} = \hat{e}q_1, q_2, q_3, q_4, q_5\ddot{e}$ are said to be equal pentagonal fuzzy numbers [31], if $p_1 = q_1, p_2 = q_2, p_3 = q_3, p_4 = q_4, p_5 = q_5$ are holds.

Definition 4. A PF-number $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5\ddot{e}$ is said to be symmetric linear pentagonal fuzzy (LPF) number $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r\ddot{e}$ when the membership [31] function of PF-number has the form

$$\mu_{\tilde{a}}(x) = \begin{cases} r\dfrac{x-p_1}{p_2-p_1}; & \text{if } p_1 \leq x \leq p_2 \\ 1-(1-r)\dfrac{x-p_2}{p_3-p_2}; & \text{if } p_2 \leq x \leq p_3 \\ 1; & \text{if } x = p_3 \\ 1-(1-r)\dfrac{p_4-x}{p_4-p_3}; & \text{if } p_3 \leq x \leq p_4 \\ r\dfrac{p_5-x}{p_5-p_4}; & \text{if } p_4 \leq x \leq p_5 \\ 0 & \text{Otherwise} \end{cases} \tag{1}$$

Definition 5. A PF-number $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5\ddot{e}$ is said to be asymmetric linear pentagonal fuzzy (LPF) number $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r, s\ddot{e}$ when the membership function of LPF-number (cf. Fig. 2) has the form

$$\mu_{\tilde{a}}(x) = \begin{cases} r\dfrac{x-p_1}{p_2-p_1}; & \text{if } p_1 \leq x \leq p_2 \\ 1-(1-r)\dfrac{x-p_2}{p_3-p_2}; & \text{if } p_2 \leq x \leq p_3 \\ 1; & \text{if } x = p_3 \\ 1-(1-s)\dfrac{p_4-x}{p_4-p_3}; & \text{if } p_3 \leq x \leq p_4 \\ s\dfrac{p_5-x}{p_5-p_4}; & \text{if } p_4 \leq x \leq p_5 \\ 0 & \text{Otherwise} \end{cases} \tag{2}$$

Remark 2.1. When $r = s$, the asymmetric LPF-number becomes symmetric LPF-number. For asymmetric LPF-number the grid value may be $r > s$ or $r < s$.

Definition 6. Let $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r, s\ddot{e}$ is said to be asymmetric linear pentagonal fuzzy (LPF) number. A α-cut set of LPF-number can be defined as $p_\alpha = \{x \in X : \mu_{\tilde{a}}(x) \geq \alpha\}$ and formulation of α-cut set are

$p_{1l}(\alpha) = p_1 + \frac{1}{r}(p_2 - p_1)$ for $\alpha \in [0; r]$ & $p_{2l}(\alpha) = p_2 + \frac{1-\alpha}{1-r}(p_3 - p_2)$ for $\alpha \in [r; 1]$

$p_{1r}(\alpha) = p_4 - \frac{1-\alpha}{1-s}(p_4 - p_3)$ for $\alpha \in [s; 1]$ & $p_{2r}(\alpha) = p_5 - \frac{1}{s}(p_5 - p_4)$ for $\alpha \in [0; s]$

where, $p_{1l}(\alpha)$ and $p_{2l}(\alpha)$ both are increasing function for $\alpha \in [0; 1]$ and $p_{1r}(\alpha)$ and $p_{2r}(\alpha)$ both are decreasing function for $\alpha \in [0; 1]$.

2.1. Arithmetic operation of two asymmetric LPF-number

Let $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r_1, s_1\ddot{e}$ and $\tilde{q} = \hat{e}q_1, q_2, q_3, q_4, q_5; r_2, s_2\ddot{e}$. The arithmetic operation of two asymmetric LPF-number can be defined as follows:

- $\tilde{a}+\tilde{q} = \hat{e}p_1+q_1, p_2+q_2, p_3+q_3, p_4+q_4, p_5+q_5; r, s\ddot{e}$ with $r \leq \max(r_1, r_2)$ and $s \leq \max(s_1, s_2)$.
- $\tilde{a}-\tilde{q} = \hat{e}p_1-q_1, p_2-q_2, p_3-q_3, p_4-q_4, p_5-q_5; r, s\ddot{e}$ with $r \leq \max(r_1, r_2)$ and $s \leq \max(s_1, s_2)$.
- Let k be any real number and $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r_1, s_1\ddot{e}$ be any asymmetric LPF-number
  If $k > 0$, then $k\tilde{a} = \hat{e}kp_1, kp_2, kp_3, kp_4, kp_5; r_1, s_1\ddot{e}$.
  If $k < 0$, then $k\tilde{a} = \hat{e}kp_5, kp_4, kp_3, kp_2, kp_1; r_1, s_1\ddot{e}$.
- $\tilde{a}\tilde{q} = \hat{e}p_1q_1, p_2q_2, p_3q_3, p_4q_4, p_5q_5; r, s\ddot{e}$ with $r \leq \max(r_1, r_2)$ and $s \leq \max(s_1, s_2)$ (Fig. 2).

3. Possibilistic middle of asymmetric LPF-number

Some concepts of the possibilistic middle of asymmetric LPF-number are introduced here to make of novel ranking method.

Definition 7. Let $p_{1l}(\alpha) = p_1 + \frac{\alpha}{r}(p_2 - p_1)$ for $\alpha \in [0; r]$ and $p_{2l}(\alpha) = p_2 + \frac{1-\alpha}{1-r}(p_3 - p_2)$ for $\alpha \in [r; 1]$ left spade of asymmetric LPF-number $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r, s\ddot{e}$. And $p_{1r}(\alpha) = p_4 - \frac{1-\alpha}{1-s}(p_4 - p_3)$ for $\alpha \in [s; 1]$ & $p_{2r}(\alpha) = p_5 - \frac{\alpha}{s}(p_5 - p_4)$ for $\alpha \in [0; r]$ right spade of asymmetric LPF-number $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r, s\ddot{e}$. The lower and upper possibilistic middle of asymmetric LPF-number can be defined as follows

$$\underline{M}(\tilde{a}) = 2\left[\int_0^r \mathrm{Pos}(\tilde{a} \leq p_{1l})\, p_{1l}\, d\alpha + \int_r^1 \mathrm{Pos}(\tilde{a} \leq p_{2l})\, p_{2l}\, d\alpha\right] \tag{3}$$

and

$$\overline{M}(\tilde{a}) = 2\left[\int_s^1 \mathrm{Pos}(\tilde{a} \geq p_{1r})\, p_{1r}\, d\alpha + \int_0^s \mathrm{Pos}(\tilde{a} \geq p_{2r})\, p_{2r}\, d\alpha\right] \tag{4}$$

where, possibilistic middle (Pos) of the events $\tilde{a} \leq p_{1l}$, $\tilde{a} \leq p_{2l}$, $\tilde{a} \geq p_{1r}$, & $\tilde{a} \geq p_{2r}$ can be calculated as

$$\mathrm{Pos}(\tilde{a} \leq p_{1l}) = \sup_{\tilde{a} \leq p_{1l}} \mu_{\tilde{a}}(x) = \alpha \quad \& \quad \mathrm{Pos}(\tilde{a} \leq p_{2l}) = \sup_{\tilde{a} \leq p_{2l}} \mu_{\tilde{a}}(x) = \alpha$$

And

$$\mathrm{Pos}(\tilde{a} \geq p_{1r}) = \sup_{\tilde{a} \geq p_{1r}} \mu_{\tilde{a}}(x) = \alpha \quad \& \quad \mathrm{Pos}(\tilde{a} \geq p_{2r}) = \sup_{\tilde{a} \geq p_{2r}} \mu_{\tilde{a}}(x) = \alpha$$

Example 3.1. Let $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r, s\ddot{e}$ be a asymmetric LPF-number. The lower and upper possibilistic middle of asymmetric LPF-number can be calculated as

$$\underline{M}(\tilde{a}) = \frac{1}{6}r^2(p_2 - p_1) + \frac{1}{6}(2r^2 - r - 1)(p_3 - p_2) + \frac{1}{2}p_2 \tag{5}$$

and

$$\overline{M}(\tilde{a}) = \frac{1}{6}s^2(p_5 - p_4) + \frac{1}{6}(2s^2 - s - 1)(p_4 - p_3) + \frac{1}{2}p_4 \tag{6}$$

Solution : Since $\tilde{a} = \hat{e}p_1, p_2, p_3, p_4, p_5; r, s\ddot{e}$ be a asymmetric LPF-number, with $r \neq s$. Now, we have to calculate the lower possibilistic middle ($\underline{M}$) and upper possibilistic middle ($\overline{M}$) of asymmetric LPF-number

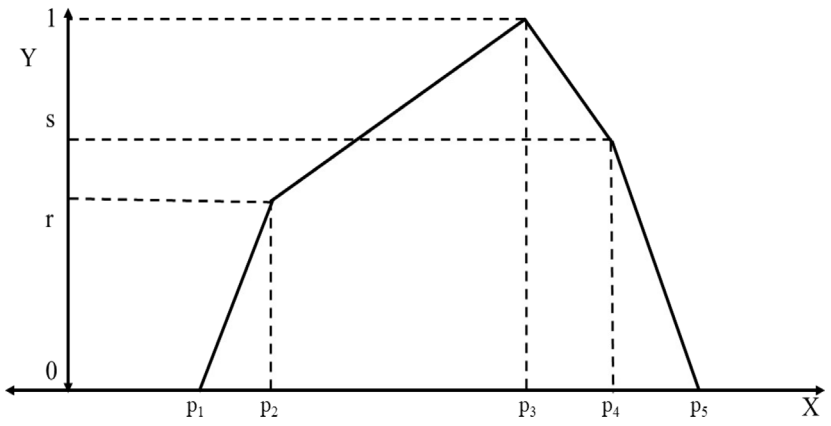

Fig. 1. Asymmetric LPF-number.

Fig. 2. Flow chart of the proposed methodology.

by the Eqs. (3) & (4).

$$\begin{aligned}\underline{M}(\phi) &= 2\left[\int_0^r Pos[\phi \le p_{1l}]\,p_{1l}\,dr + \int_r^1 Pos[\phi \le p_{2l}]\,p_{2l}\,dr\right]\\ &= 2\left[\int_0^r \left(p_1 + \frac{r}{r}(p_2-p_1)\right)dr + \int_r^1 \left(p_2 + \frac{1-r}{1-r}(p_3-p_2)\right)dr\right]\\ &= 2\left[\int_0^r \left(p_1 + \frac{r}{r}(p_2-p_1)\right)dr + \int_0^1 \left(p_2 + \frac{r}{1-r}(p_3-p_2)\right)dr\right]\\ &= \frac{1}{6}r^2(p_2-p_1) + \frac{1}{6}(2r^2-r-1)(p_3-p_2) + \frac{1}{2}p_2\end{aligned}$$

And

$$\begin{aligned}\overline{M}(\phi) &= 2\left[\int_0^s Pos[\phi \ge p_{1r}]\,p_{1r}\,ds + \int_s^1 Pos[\phi \ge p_{2r}]\,p_{2r}\,ds\right]\\ &= 2\left[\int_0^s \left(p_4 - \frac{1-s}{1-s}(p_4-p_3)\right)ds + \int_s^1 \left(p_5 - \frac{s}{s}(p_5-p_4)\right)ds\right]\\ &= 2\left[\int_0^1 \left(p_4 - \frac{1-s}{1-s}(p_4-p_3)\right)ds + \int_0^s \left(p_5 - \frac{s}{s}(p_5-p_4)\right)ds\right]\\ &= \frac{1}{6}s^2(p_5-p_4) + \frac{1}{6}(2s^2-s-1)(p_4-p_3) + \frac{1}{2}p_4\end{aligned}$$

Note: When $r = s$, the possibilistic middle ($M(\phi)$) of symmetric LPF-number $\phi = \hat{e}p_1;p_2;p_3;p_4;p_5;r\ddot{e}$ can be defined as

$$\begin{aligned}M(\phi) &= \frac{\overline{M}(\phi) + \overline{M}(\phi)}{2}\\ &= \frac{1}{6}r^2(p_2+p_5-p_1-p_4) + \frac{1}{6}(2r^2-r-1)(p_4-p_2) + \frac{1}{2}(p_2+p_4)\end{aligned} \quad (7)$$

3.1. Ranking method of symmetric LPF-number based on possibilistic middle

Let $\phi = \hat{e}p_1;p_2;p_3;p_4;p_5;r_1\ddot{e}$ and $\varphi = \hat{e}q_1;q_2;q_3;q_4;q_5;r_2\ddot{e}$ be two symmetric LPF-number. The possibilistic middle-based ranking method between two symmetric LPF numbers can be defined as follows:

(i) If $M(\phi) > M(\varphi)$, then ϕ is bigger than φ, its noted by $\phi \succ \varphi$.
(ii) If $M(\phi) < M(\varphi)$, then ϕ is smaller than φ, its noted by $\phi \prec \varphi$.
(iii) If $M(\phi) = M(\varphi)$, then ϕ is equivalent to φ, its noted by $\phi = \varphi$.

4. Possibilistic middle-based MCDM method under symmetric LPF-number

In this section, we have defined a multi-criteria decision-making method based on possibilistic middle. Here, the information of the decision maker considers a symmetric LPF number. Let $U = \{U_1;U_2;U_3;\dots;U_n\}$ sets of n-criteria and $V = \{V_1;V_2;V_3;\dots;V_m\}$ be the sets of m-alternatives of the decision making problem. Every criterion has a different characteristic. Therefore we have chosen the n-weight vector $w = \{w_1;w_2;\dots;w_n\}$ of the decision-making problem with $w_j > 0$ and $\sum_{j=1}^{n} w_j = 1$. For the proposed decision-making problem, we